Part1 *

Trim Entities | Convert Entities | Offset Entities | Offset On Surface | Mirror Entities | Linear Sketch Pattern | Move Entities | Add Relation | Repair Sketch | Quick Snaps | Rapid Sketch | Instant2D | Shaded Sketch Contours

valuate | Simulation | Analysis Preparation

Material

Search...

- SOLIDWORKS Materials
  - Steel
    - 1023 Carbon Steel Sheet (SS)
    - 201 Annealed Stainless Steel (SS)
    - A286 Iron Base Superalloy
    - AISI 1010 Steel, hot rolled bar
    - AISI 1015 Steel, Cold Drawn (SS)
    - AISI 1020
    - AISI 1020 Steel, Cold Rolled
    - **AISI 1035 Steel (SS)**
    - AISI 1045 Steel, cold drawn
    - AISI 304
    - AISI 316 Annealed Stainless Steel Bar
    - AISI 316 Stainless Steel Sheet (SS)
    - AISI 321 Annealed Stainless Steel (SS)
    - AISI 347 Annealed Stainless Steel (SS)
    - AISI 4130 Steel, annealed at 865C
    - AISI 4130 Steel, normalized at 870C
    - AISI 4340 Steel, annealed
    - AISI 4340 Steel, normalized
    - AISI Type 316L stainless steel

Access more materials from
SOLIDWORKS Materials Web Portal

Add...

Properties | Tables & Curves | Appearance | CrossHatch | Custom | Application Data

Material properties
Materials in the default library can not be edited. You must first copy the material to a custom library to edit it.

Model Type: Plasticity - von Mises

Save model type in library

Units: SI - N/mm^2 (MPa)

Category: Steel

Create stress-strain curve

Name: AISI 1035 Steel (SS)

Default failure criterion: Max von Mises Stress

Description:

Source:

Sustainability: Defined

Property	Value	Units
Elastic Modulus	204999.9984	N/mm^2
Poisson's Ratio	0.29	N/A
Tensile Strength	585.0000029	N/mm^2
Yield Strength	282.685049	N/mm^2
Tangent Modulus		N/mm^2
Thermal Expansion Coefficient	1.1e-05	/K
Mass Density	7849.999987	kg/m^3
Hardening Factor	0.85	N/A

Save | Config... | Apply | Close | Help

5 SP2.1 - Instructional Use Only | Editing Part | Simplified Interface

Search